

1) Diket

$$t_i = 205$$

$$h = 105$$

$$T(20) = 68.5^\circ\text{C}$$

$$T(30) = 61.9^\circ\text{C}$$

nilai turunan bernilai negatif = suhu

benda berkurang thdp waktu sehingga

benda mengalami proses pendinginan

Formus forward Difference:

$$\left. \frac{dT}{dt} \right|_{t=t_i} = \frac{T(t_i + h) - T(t_i)}{h}$$

Subst:

$$\left. \frac{dT}{dt} \right|_{t=20} = \frac{61.9 - 68.5}{10} = \frac{-6.6}{10} = -0.66^\circ\text{C/s}$$

2) Diket

$$t_i = 1.0$$

$$h = 0.25$$

$$V(1.0) = 1.11\text{ V}$$

$$V(0.8) = 1.50\text{ V}$$

Tanda negatif menunjukkan

kapasitor sedang melepaskan muatan

sehingga arah arus berlawanan

dengan arah pengisian.

Formus Backward Difference:

$$\left. \frac{dV}{dt} \right|_{t=t_i} = \frac{V(t_i) - V(t_i - h)}{h}$$

Subst:

$$\left. \frac{dV}{dt} \right|_{t=1.0} = \frac{1.11 - 1.50}{0.2} = \frac{-0.39}{0.2} = -1.95\text{ V/s}$$

menghitung arus

$$\text{Diket: } C = 470\text{ pF}$$

$$C = 470 \times 10^{-6}\text{ F}$$

$$I = C \frac{dV}{dt}$$

$$I = (470 \times 10^{-6}) (-1.95)$$

$$= -9.165 \times 10^{-9}\text{ A}$$

$$= -0.9165\text{ nA}$$

3) Diket

$$t_i = 1,55$$

$$h = 0,55$$

$$x(2,0) = 3,92 \text{ m}$$

$$x(1,0) = 1,48 \text{ m}$$

kecepatan bernilai positif, sehingga partikel bergerak ke arah sumbu x positif dan posisinya semakin bertambah terhadap waktu.

Formula Central Difference:

$$\frac{dx}{dt} \bigg|_{t=t_i} = \frac{x(t_i + h) - x(t_i - h)}{2h}$$

Substitusi:

$$v(1,5) = \frac{3,92 - 1,48}{2(0,5)} = \frac{2,44}{1} = 2,44 \text{ m/s}$$